

Section 2 – Shear and Moment Diagrams

Provide information for constructing the **shear diagram**. The support reactions are $A_y = 2300$ N acting up, $B_y = 2600$ N acting up. A is a pin and B is a roller.

1. (3 pt) What happens to the shear diagram at $x = 6$ m?

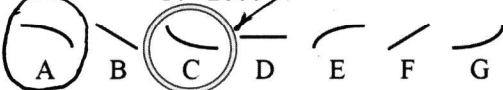
✓ A. Drop 500 N D. Drop 400 N
B. Jump 2300 N E. Jump 500 N
C. Jump 2600 N F. Drop 800 N

What is the numerical value (magnitude only) of the change in shear for the region $0 < x < 6$ m and what is the behavior (shape) of the shear diagram for the region $0 < x < 6$ m?

2. (3 pt) Change = _____ (magnitude only)

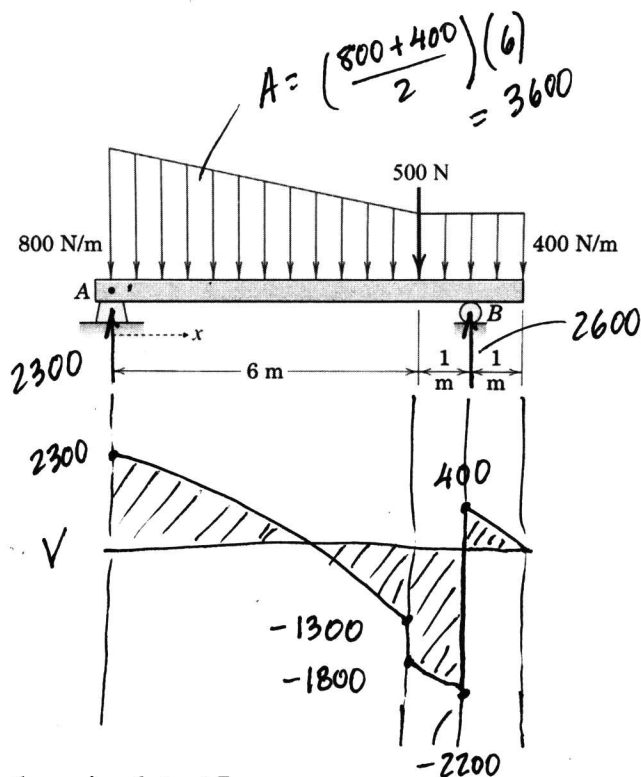
✓ A. 3600 N D. 2400 N
B. 2300 N E. 500 N
C. 1200 N F. 2600 N

3. (3 pt) Behavior:



4. (3 pt) Determine the order of the equation for the internal shear for the region $6 < x < 7$ m.

✓ A. first order, linear D. second order, parabolic
B. zero E. third order cubic
C. constant



Provide information for constructing the **moment diagram**. The support reactions are $A_y = 8$ kip acting down, $M_A = 60$ kip-ft acting clockwise. A is a fixed support. The shear diagram is provided.

What is the numerical value (magnitude only) of the change in moment for the region $5 < x < 7$ ft and what is the behavior (shape) of the moment for the region $5 < x < 7$ ft?

5. (3 pt) Change = _____ (magnitude only)

✓ A. 8 kip-ft D. 12 kip-ft
B. 5 kip-ft E. 30 kip-ft
C. 20 kip-ft F. 4 kip-ft

6. (3 pt) Behavior:

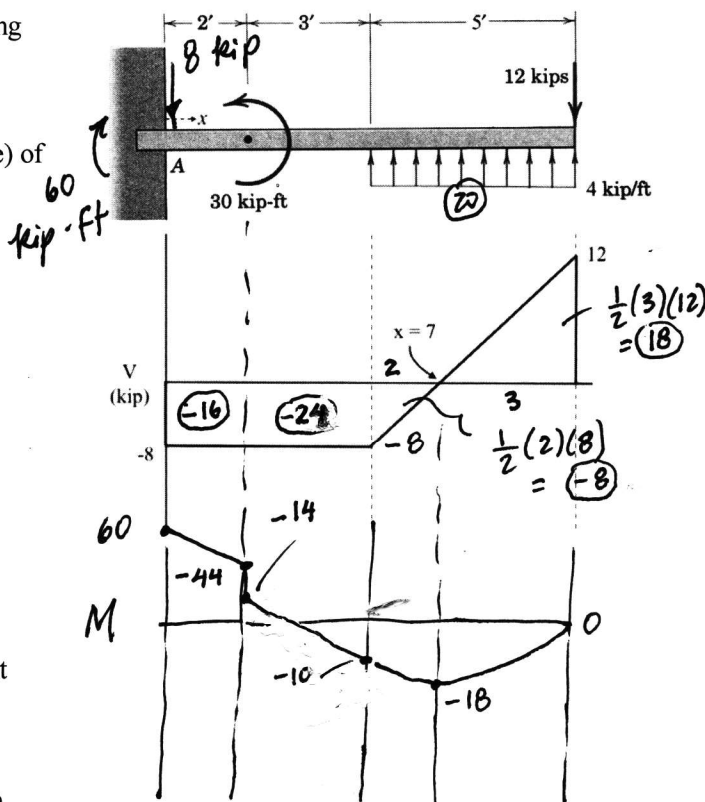


7. (3 pt) What happens to the moment diagram at $x = 2$ ft?

✓ A. Drop 30 kip-ft D. Drop 8 kip-ft
B. Drop 16 kip-ft E. Jump 16 kip-ft
C. Jump 30 kip-ft F. Drop 60 kip-ft

8. (4 pt) Determine the equation for the internal bending moment for the region $2 < x < 5$ ft.

✓ A. $-8x + 30$
B. $-8x + 60$
C. $8x^2 - 16x$
D. $8x^2 + 24x$
E. $40x$



Handwritten calculations for the internal bending moment:

$$2 \leq x \leq 5$$

$$60 \uparrow \quad \downarrow 8 \quad \downarrow 30 \quad \uparrow M$$

$$1 - x \quad \uparrow V$$

$$+ \uparrow -V - 8 = 0 \quad \downarrow$$

$$V = -8 \text{ kip}$$

$$M + 30 + 8x - 60 = 0$$

$$M = 30 - 8x$$